sprintd Zurich 2026: Equans case playbook

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Event overview

We are doing the Equans case at sprintd Zurich: 36 hours at Bluelion Zurich, Friday 9 Oct 17:00 to Sunday 11 Oct, about 100 builders. It is the organizer's first branded sprint (sprintd GmbH, founded by ex START Hack people).

Item	Detail
Case partners	Equans (tender pricing), Inventx with Outcept (Swiss German call compliance)
Supporting partners	OpenAI, Anthropic, ElevenLabs, n8n, Exoscale (credits not confirmed)
Case pitches	Sunday 09:00 to 11:00: 5 min demo, 3 min Q&A, 2 min change over
Equans case winner	Announced by 12:00 Sunday
Grande Finale	Sunday 13:00 to 14:30, neutral jury, overall prize
Overall prize pool	From CHF 2,000
Equans case prize	Up to 2 internship, direct entry or paid follow up spots, gift packages for the top 3 teams

Partners own the solutions and judge their own case, and both pitches stressed continuing after the weekend. A credible "this runs on Monday" story earns points.

The Equans case: from tender to priced offer

Build a tool that takes the Schulhaus Oberglatt ventilation tender, finds the LV positions, matches them to the catalogue and benchmarks, and proposes quantities, install hours and a first price with every assumption visible.

Criterion	Weight	What it means for us
Impact and relevance	30%	A price an estimator would trust

Criterion	Weight	What it means for us
Feasibility	25%	Azure capable, fits the bid process (Azure and OpenAI are a plus, not required)
Traceability	20%	Every number linked back to the tender page or reference row
Design and usability	15%	Usable without training
Creativity and innovation	10%	

Must have: a working prototype on the provided data, tender package to offer, every number traceable, plus a few sentences or one slide on the path to production. **Stretch:** bid optimization (pricing scenarios vs margin, competitiveness, positioning). Slides are optional; the demo matters more.

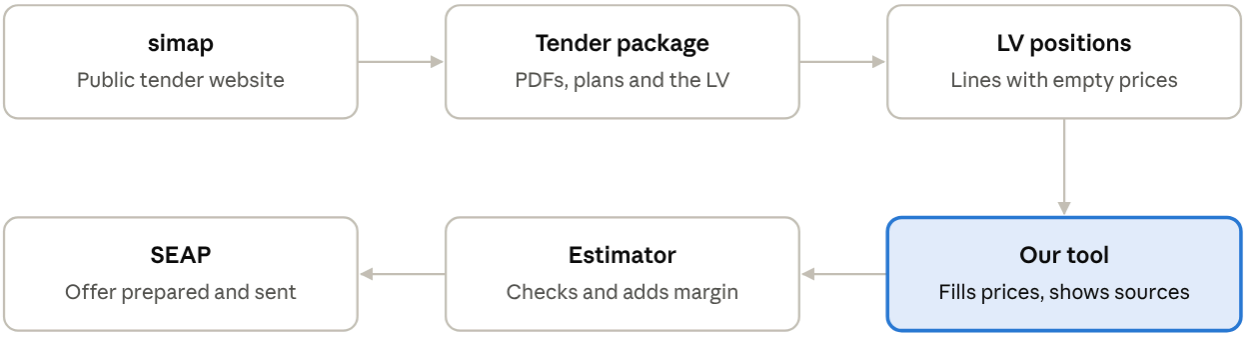
What the Friday Q&A added

- They are "not looking for a perfect price model". The process working end to end matters more than accuracy, and prices stay adjustable.
- No price forecasting needed: price swings are already covered by margins and the project parameter factors.
- For this case every material is in the catalogue. In real life they email suppliers for missing items and type the quote in by hand, which "happens a lot". A supplier quote loop (flag missing item, draft request, read reply, fill price) earns extra points.
- Bid optimization as they see it: more competitors means a lower margin but a better chance to win. Today that is the estimator's gut feel; they want the tool to learn from past tenders and suggest the approach with the best win chance that is still viable.
- Other ideas they named: learning from history, alternative product suggestions (sustainability is a push at Equans).
- The presenter is not from building technology either, so clarity beats domain depth.

Key concepts

The tender comes from simap, the LV is the part we price, and SEAP is where Equans assembles the offer. Only the pricing step is ours to build.

Only the pricing step is ours to build



tender to offer flow · 6 steps

Term	Plain meaning
simap	Switzerland's official public tender website. Confederation, cantons and communes publish calls for tender there (above CHF 230,000 for supplies or services). Firms download the documents and ask questions in a forum. We do not need to connect to it.
SEAP	The platform where Equans bid teams prepare public procurement offers. Not a public site we could find; likely internal or third party. No integration required, so our output should be easy to carry into it (clean table, Excel).
LV (Leistungsverzeichnis)	Bill of quantities: the list of everything the client wants built, one line per item.
LV position	One priced line: text, quantity, unit. The client fills quantity, the bidder fills unit price and total.
NPK	CRB's standard Swiss position catalogue. Ventilation sits in chapters 461 to 463 (464 controls), but NPK is not yet standard in ventilation, so real LVs are mostly free text.
Hauptposition / Unterposition	Main heading (text only) and the priced sub line under it. A sub line often only makes sense with its parent text, e.g. "Same, Ø 315 mm".
BKP 244	Swiss building cost code for ventilation systems (Lüftungsanlagen).

Term	Plain meaning
SIA interface	The standard Swiss file format for LVs. The Oberglatt client requires it with the offer.

The story: the commune of Oberglatt posts the tender on simap, Equans downloads the package, an estimator prices each LV position (today by Ctrl+F through handbooks), adds margin, and submits the offer prepared in SEAP.

Starter package

The maths is solved; reading the 202 page PDF into position rows is the real work.

File	What it holds
BKP_244_LV_Lueftung_260804.pdf	Full tender, 202 pages, German, selectable text (no OCR needed)
catalog.csv	56 products: list price, discount, net cost, install_code, catalogue vs supplier package, supplier_quote_required
productivity.csv	22 install rules: base hours per unit, labour role, hourly rate (96, 118, 145, 155 CHF), which factors apply
project_parameters.json	Productivity factors, allowances, project costs, overhead 8.5%, margin 5.5%, bid optimization settings
example_lv_positions.csv	22 verified positions with page, quantity, product and install code (ground truth)
expected_output_example.json	Golden numbers for those 22, the subset total and 3 bid scenarios

How the PDF is laid out

- Pages 1 to 66: conditions, award criteria, schedule. Not priceable.
- Pages 67 to 199: the LV, by system: LA01 Nebenräume UG, LA02 Essenräume EG, LA03 Küche/Abwaschen, LA04 Schulzimmer OG, LA05 Turnhalle (Los 2), 244.06 Montagesystem Flachdach.
- Each system has sections A Apparate, B Kanäle, C Armaturen, D Regulierung, E Montage, F Dämmungen.

Parsing traps

- No position numbers. A position is a free text block ending in unit + quantity, e.g. "Stk 5".
- One block can hold several sizes with their own quantities. Page 78: Ø 10 cm Stk 5 and Ø 12.5 cm Stk 6 become LV0001 and LV0002.
- A simple regex finds about 200 quantity lines (Stk 169, m2 26, m 3).
- LA05 Turnhalle puts the quantity at the start of the line and needs its own rule, plus the 1.25 sports hall mounting factor.
- Lines with a unit but no quantity get status insufficient_information.
- Lump sum lines ("Total … .....") and referenced supplier offers ("Offerte P241783") appear too.
- The catalogue is small, so most of the full tender will not match. Never force a price: use the README statuses calculated, supplier_quote_required, assumption_applied, review_required, insufficient_information, out_of_scope, non_priceable.

How the parser spots a position

The PDF text is read line by line. A line that ends in a unit and a number closes a position; the lines above it, back to the previous position or heading, are its description. Page 169 (LV0017):

```
Thermische Aussendämmung 60/10                <- description
aus 60 mm Mineralwollplatten ( = 80 kg/m3)    <- description
mit Alu-Folie, gestiftet           m2 12      <- unit + quantity: position
ends here
```

Result: 60 mm insulation, 12 m2, page 169. In code this is a pattern such as `(Stk|m2|m|kg|h|psch)\s+(\d+)$` at the end of a line. When a block has several such lines (Ø 10 cm Stk 5, Ø 12.5 cm Stk 6), each becomes its own position sharing the block's description.

Hidden gems on page 9

- Award criteria: price 60 points (lowest bid 100, 150% of lowest 0, linear), quality 25, capacity 10, apprentices 5. These are points, not percent: the client scores every bid out of 100 points and price can earn up to 60 of them. Example: the lowest bid is 400k and gets all 60. A bid of 420k is 5% higher, so it loses 5/50 of the price points: 90 of 100 × 0.6 = 54 points. Every 1% above the lowest bid costs 1.2 points, which must be won back on quality (25), capacity (10) or apprentices (5). A real model for bid optimization.
- Submission must include the LV as a SIA interface file. Ready made line for the path to production note.

Pricing maths

pricing_engine.py reproduces every golden number: 22 of 22 positions, the subset baseline of CHF 414,489.12 and all 3 bid scenarios. The LLM only picks product_id and install_code; plain code does the rest.

What pricing_engine.py is. A short Python file in the Project: the Compute step. Give it a position (quantity, product_id, install_code) and it reads catalog.csv, productivity.csv and project_parameters.json, then returns material, hours, labour, total, status and an explain trail with every operand. It also computes the project baseline and the 3 bid scenarios. Run on the package, it prints 22/22 matches against expected_output_example.json. It does not read the PDF or choose products: the parser and matcher feed it.

Per position

```
\text{material} = q \times \text{net cost} \times 1.04
```

```
\text{hours} = q \times \text{base hours} \times \tfrac{1}{0.88} \times \prod \text{other listed factors}
```

```
\text{labour} = \text{hours} \times \text{hourly rate}, \quad \text{direct} = \text{material} + \text{labour}
```

Factors: site efficiency 0.88 (inverted), school logistics 1.07, mounting height school 1.1 or sports hall 1.25, timber interface 1.08. Each install rule lists which ones apply.

Worked example, LV0001 (5 supply air valves DN100, page 78, P-AIR-001, DIFFUSER)

Step	Calculation	CHF
Material	5 × 74.80 × 1.04	388.96
Hours	5 × 0.65 × 1.1364 × 1.1 × 1.07 = 4.35 h	
Labour	4.347 h × 96	417.30
Direct	388.96 + 417.30	806.26

Project and bid

```
\text{baseline} = \text{direct} \times 1.025 \times 1.085
```

```
\text{bid} = \lceil \text{baseline} / (1 - m) \rceil_{1000}
```

$$P(\text{win}) = 0.55 - 0.05 \times \%\text{ above the price at } 5.5\% \text{ margin}, \quad \text{clamped } 0.05 \ldots 0.95$$

$$\text{expected contribution} = P(\text{win}) \times (\text{bid} - \text{baseline})$$

The 0.55 is reference 0.55 + strategic appetite 0.05 + competition −0.05.

Scenario	Margin	Bid (CHF)	Win probability	Expected contribution (CHF)
Aggressive	2.5%	426,000	0.69	7,986
Target	4.0%	432,000	0.63	10,951
Conservative	6.5%	444,000	0.49	14,419

These are for the 22 position golden subset only (direct CHF 372,700). The full tender total needs the open question in the questions section answered.

Solution design

Build a glass box estimator: the LLM picks, code computes, the estimator decides. In a 2026 University of Ljubljana study, recent models (GPT-5, Claude Sonnet 4.5) drafting quotes from customer emails picked the right items about 50% of the time and got quantities right at most 15%. Their task was harder than ours (quantities had to be inferred; ours are printed in the LV), but the lesson holds: let the LLM choose from a short list, and keep every number in code, where it is exact, repeatable and traceable.

Build for any tender, not just Oberglatt. Equans said it in the intro ("not only for the test data but for all tenders"), and the README calls the 22 verified positions ground truth for testing and judging, while the prototype must attempt the full tender.

- Nothing Oberglatt specific in code: no hardcoded page ranges, system names, product IDs or example rows. Read systems and sections from the PDF's own headings.
- Catalogue, productivity rules and project parameters are inputs that change per project. pricing_engine.py already reads them from files.
- The parser targets patterns, not this PDF: quantity at line end or line start, several sizes in one block, missing quantities, lump sums.
- Anything unknown gets a status, never a forced match. A tender full of products outside the catalogue must still give a reviewable result.
- The 22 examples are a test set, not something to tune against line by line. Report accuracy on them.

- Demo proof: let the jury pick any LV page and price it live.

Option	What it is	Extra effort	Verdict
A. Glass box pipeline	Parse PDF to positions with page refs, hybrid match (keyword + embeddings + LLM picks 1 of top 5 catalogue IDs or none), pricing_engine.py computes, review grid with PDF highlight and an explain drawer	Core, about 40 to 50 person hours	Build
B. Excel output	Workbook with live formulas pointing at catalogue and parameter sheets, cell comments citing source	Low	Build as the export
C. Chat copilot	"Why is this line so expensive?" over A's API	8 to 10 h	Only if time remains
D. Learning loop	Estimator corrections reused on the next tender, anomaly flags	6 to 10 h	Nice to have
E. Bid optimizer	Recommended Bid Range: 3 scenarios, competitor and margin sliders, win probability, page 9 scoring model	6 to 8 h	Build (stretch, named by Equans)
F. Supplier quote loop	Flag supplier_quote_required items, draft the request email, read reply, fill the price	4 to 6 h	Build (extra points in Q&A)

Recommended build: A to F (all of them)

How the top 5 candidates are picked

1. Normalize: the LLM (or rules) turns the German block into English attributes. "Zuluft Tellerventil, Kunststoff, Ø 10 cm, Stk 5" becomes supply air valve, DN100, pc. Unit conversion (Ø 10 cm = DN100) happens in code.
2. Shortlist in code: drop catalogue items with the wrong unit, score the rest by word overlap and/or embedding similarity, keep the best 5.
3. Pick: the LLM sees the position and the 5 candidates and must answer one product_id from that list or "none", plus a one line reason. It cannot invent an ID.

LV0001 shortlist from a simple word overlap run on catalog.csv:

Rank	Candidate	Score	Why it is there
1	P-AIR-001 Supply air valve DN100	0.57	Right type, right size
2	P-AIR-002 Supply air valve DN125	0.57	Right type, wrong size
3	P-AIR-003 Exhaust air valve DN100	0.43	Right size, but exhaust not supply
4	P-FIR-001 Fire damper EI90-S DN100	0.14	Weak match
5	P-DAM-001 Manual balancing damper DN125	0.14	Weak match

Ranks 1 and 2 tie on words alone, which is exactly why the size check and the LLM pick exist. The LLM answers P-AIR-001 ("supply air, plastic, Ø 10 cm = DN100"), the same as Equans' verified answer.

Trust features

1. Explain drawer: tender text with page link, matched catalogue row plus runner ups, formula with every operand linked to its source.
2. Status per position (the 7 README statuses) and a confidence light, review queue sorted by value at risk.
3. Coverage line at the top, e.g. "62% calculated, 18% supplier quote, 20% needs you".
4. Totals by system (LA01 to LA05) and by category.
5. Estimator can swap a match or an assumption and the total updates.
6. Provider swappable: Azure OpenAI for the demo (feasibility points), a Swiss hosted model for production.

Team split (4 people)

- P1: PDF parser to positions (system, section, text, qty, unit, page).
- P2: Matching and LLM judge, accuracy against the 22 verified positions.
- P3: Review UI (grid, PDF viewer, explain drawer).
- P4: Pricing engine integration, export, bid optimizer, supplier quote loop, pitch.

Demo script (5 minutes)

1. "Estimators spend hours on Ctrl+F. Watch." Upload the tender PDF.
2. Positions fill system by system; coverage line shows the split.
3. Click LV0001: PDF jumps to page 78, drawer shows 5 × 74.80 × 1.04 and the hours formula. 806.26 CHF matches Equans' own number.
4. Open a supplier quote item: draft email ready, paste a reply, price fills in.

5. Swap a match; the total updates.
6. Recommended Bid Range: move the competitor slider, watch margin and win probability shift.
7. Export to Excel with live formulas.
8. Close: "LLM picks, code computes, estimator decides. Runs on Azure; SIA export is the next step."

Open questions for Equans

- [ ] For the full tender total, do we add project costs (engineering 420 h at 145, commissioning 160 h at 155, hygiene inspection 18,000, documentation 12,000), small material 3.5% and supplier contingency 3%? The golden subset uses none of them.
- [ ] How should the sports hall (Los 2) be priced: separate total, same overhead and margin?
- [x] ~~Should the UI be in German, English or both?~~
  - [x] ~~Answer: both~~
- [ ] Do real Equans tenders come with NPK numbers or as free text? We plan free text matching plus an optional NPK code field that is matched exactly when present. Is that enough, or should NPK matching be in the demo?

Sources and files

Sources

- sprintd case page: From tender to priced offer
- sprintd events and partners
- SIMAP: the Swiss public procurement platform (SME Portal)
- CRB NPK user guide 2025
- EC3 2026 study on LLM quotation drafting
- Starter package: Schulhaus Oberglatt Hackathon Starter Package v2 (README, tender PDF pages 8, 9, 78)
- Our recordings of the Friday intro presentation and the Equans Q&A

Files in the Project

- pricing_engine.py: deterministic engine, verified against all golden outputs
- starter package notes.md
- sprintd hackathon case options.md: full research report on both cases
- intro presentation transcript.txt and equans QA transcript.txt